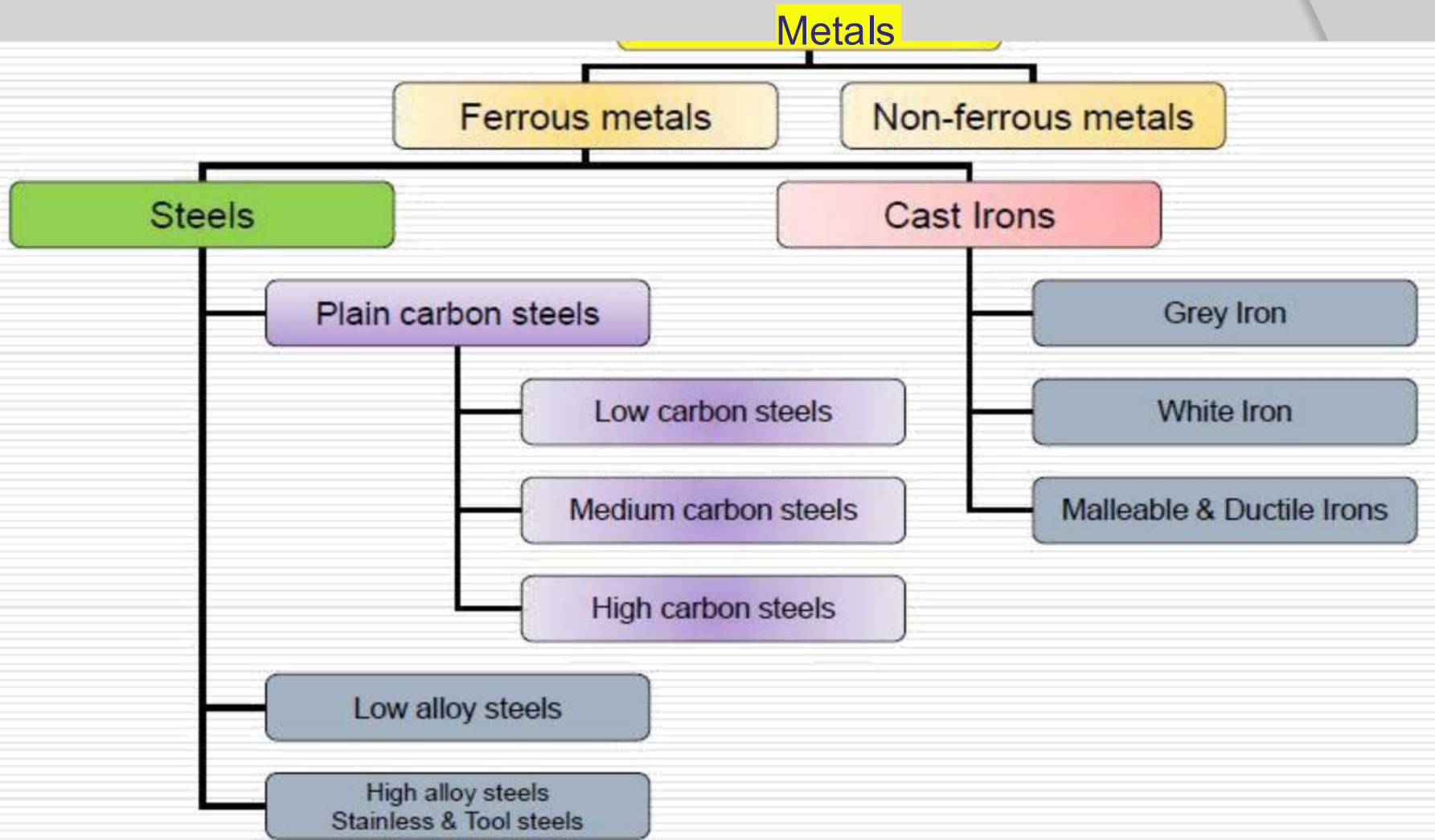


Material Science & Engineering

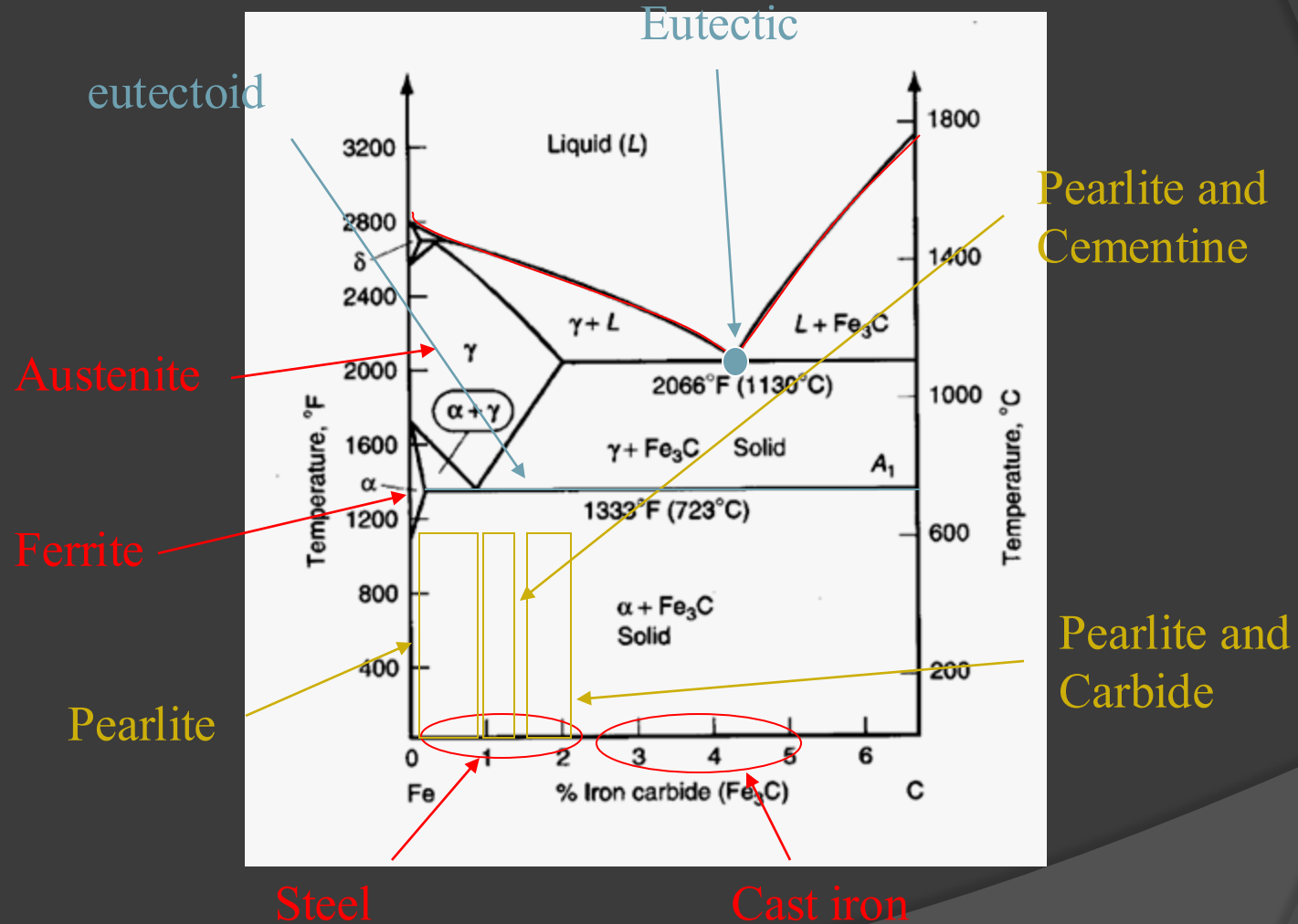
Outline

- Introduction
- Cooling curve for pure iron
- Definition of structures
- Iron-Carbon equilibrium phase diagram – Sketch
- **The Iron-Iron Carbide Diagram - Explanation**
- The Austenite to ferrite / cementite transformation
- Nucleation & growth of pearlite
- Effect of C %age on the microstructure of steel
- Relationship b/w C %age & mechanical properties of steel

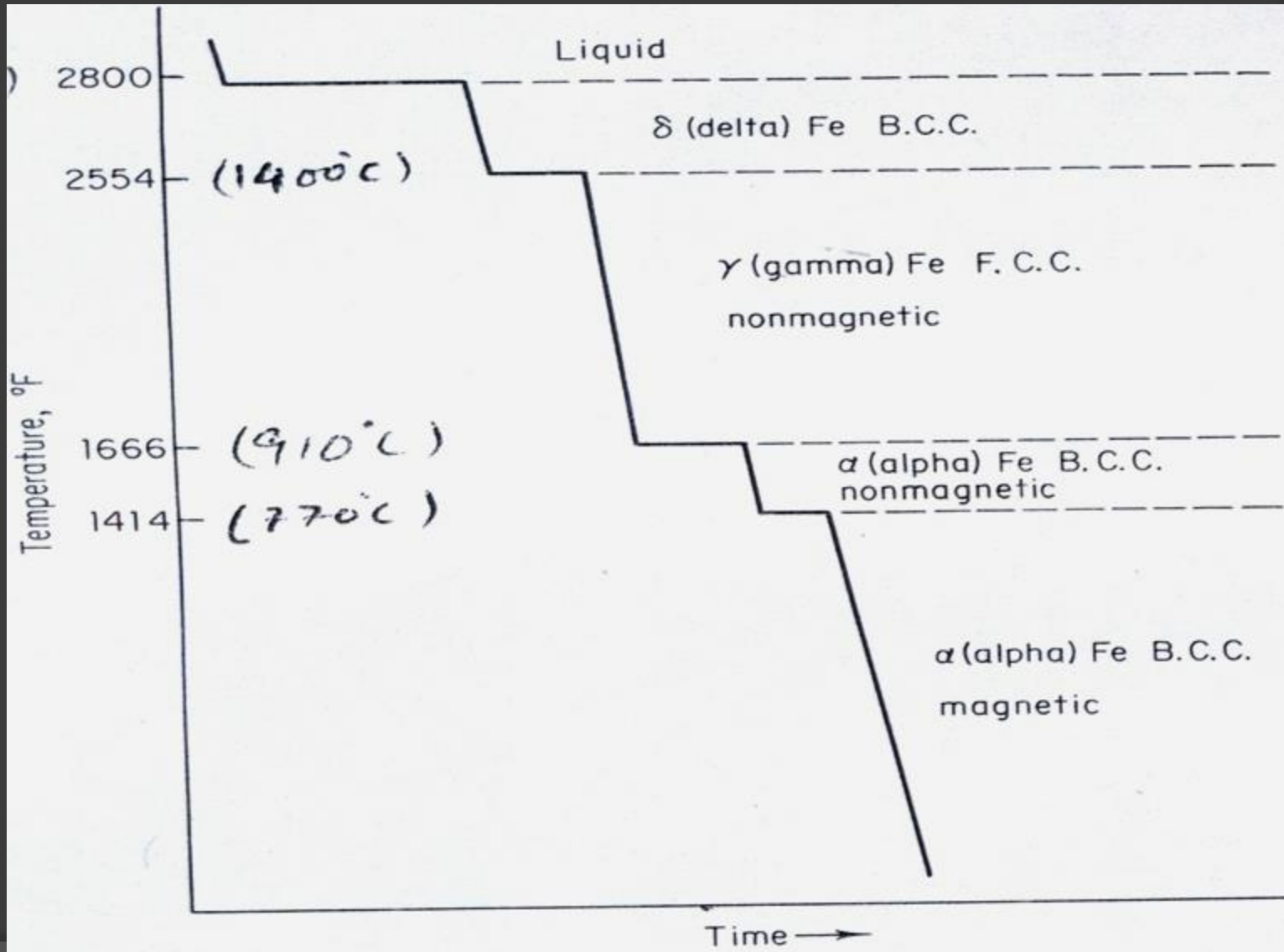
Fe-C alloy classification



IRON-CARBON PHASE DIAGRAM



Cooling curve for pure iron

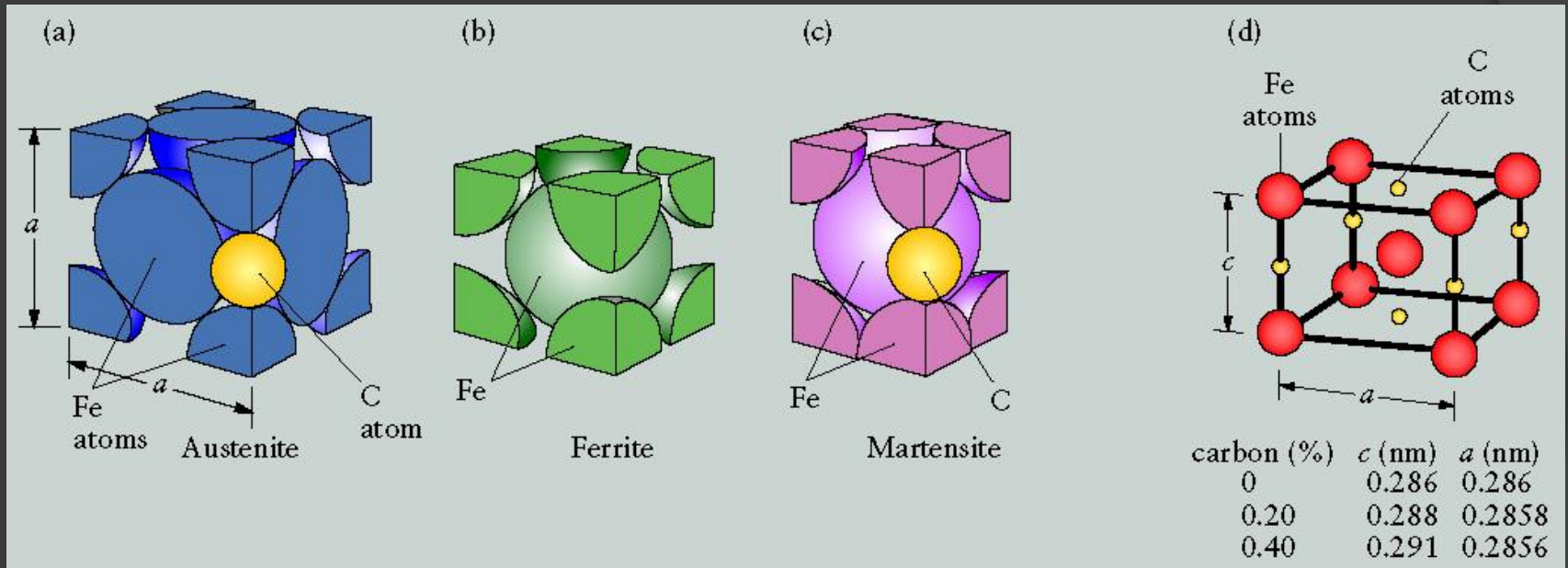


Definition of structures

Various phases that appear on the Iron-Carbon equilibrium phase diagram are as under:

- Austenite
- Ferrite
- Pearlite
- Cementite
- Martensite*
- Ledeburite

Unit Cells of Various Metals



- FIGURE - The unit cell for (a) austenite, (b) ferrite, and (c) martensite. The effect of the percentage of carbon (by weight) on the lattice dimensions for martensite is shown in (d). Note the interstitial position of the carbon atoms and the increase in dimension c with increasing carbon content. Thus, the unit cell of martensite is in the shape of a rectangular prism.

Microstructure of different phases of steel

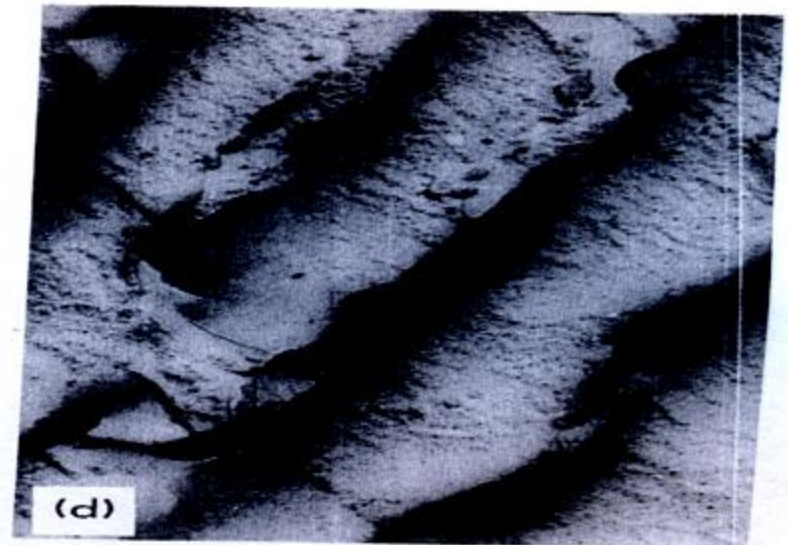
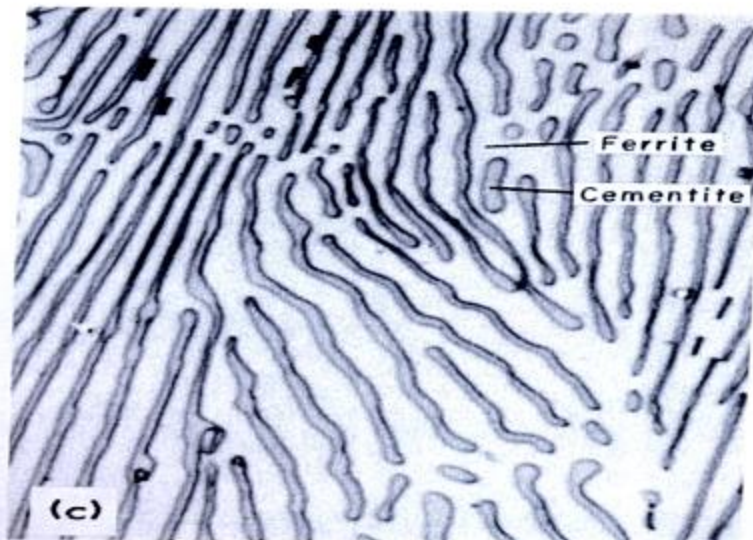
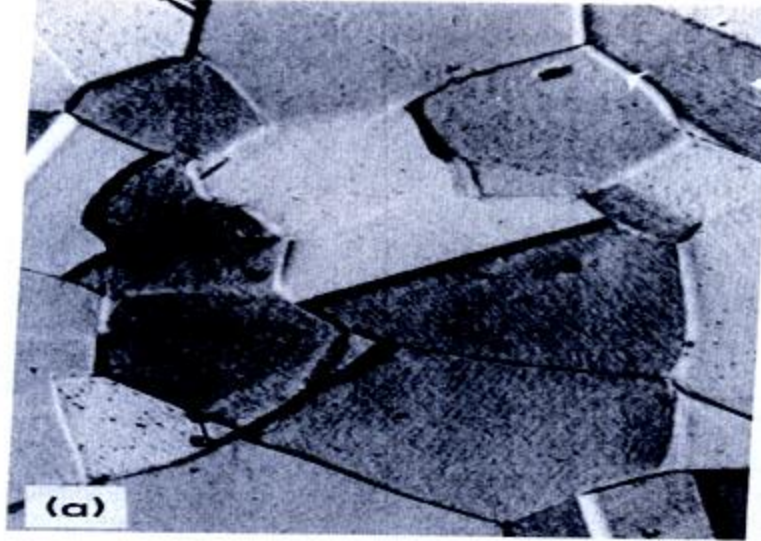


Fig. 7-8 The microstructure of (a) austenite, 500X; (b) ferrite, 100X; (c) pearlite, 2,500X; (d) pearlite, electron micrograph, 17,000X; enlarged 3X in printing. (a, b, and c, Research Laboratory, U.S. Steel Corporation.)

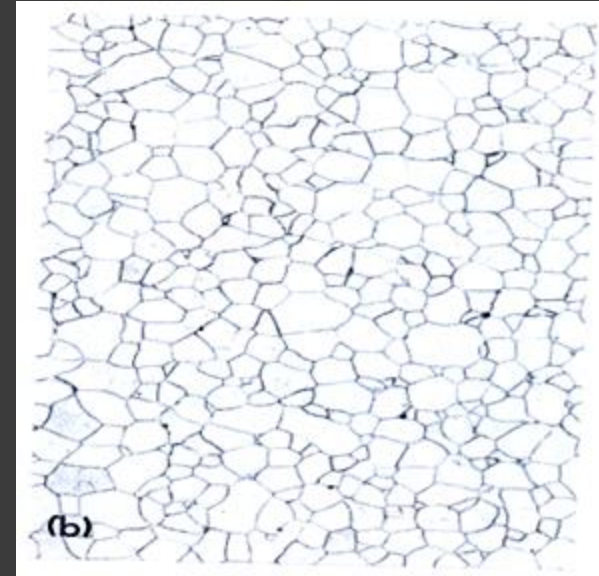
Definition of structures

- **Ferrite** is known as α solid solution.
- It is an interstitial solid solution of a small amount of carbon dissolved in α (BCC) iron.
- stable form of iron below 912 deg.C
- The maximum solubility is 0.025 % C at 723°C and it dissolves only 0.008 % C at room temperature.
- It is the softest structure that appears on the diagram.

Definition of structures

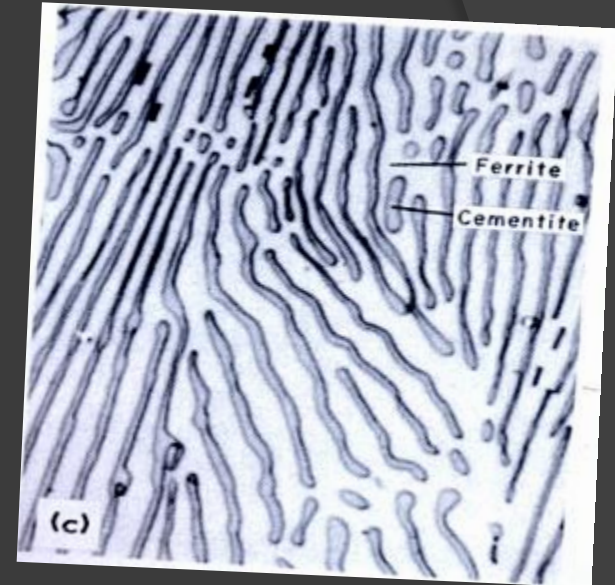
Ferrite

- ⦿ Average properties are:
 - Tensile strength = 40,000 psi;
 - Elongation = 40 % in 2 in;
 - Hardness > Rockwell C 0 or
> Rockwell B 90



Definition of structures

- **Pearlite** is the eutectoid mixture containing 0.80 % C and is formed at 723°C on very slow cooling.
- It is a very fine platelike or lamellar mixture of ferrite and cementite.
- The white ferritic background or matrix contains thin plates of cementite (dark).



Definition of structures

Pearlite

⦿ Average properties are:

- Tensile strength = 120,000 psi;
- Elongation = 20 % in 2 in.;
- Hardness = Rockwell C 20, Rockwell B 95-100, or BHN 250-300.

Definition of structures

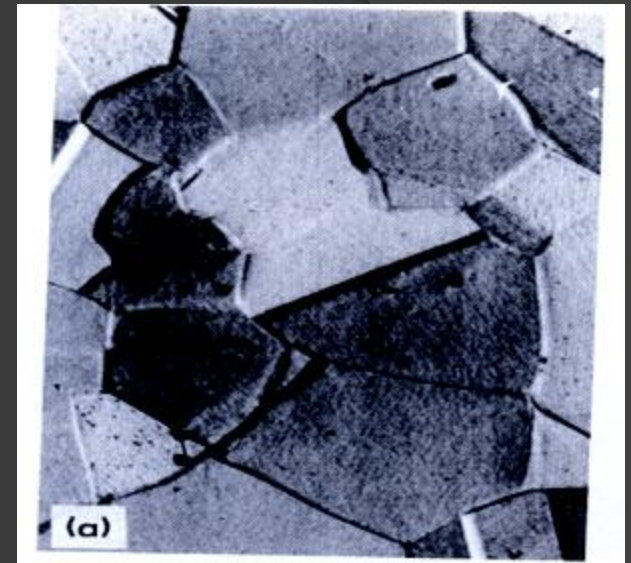
- ⦿ **Austenite** is an interstitial solid solution of Carbon dissolved in γ (F.C.C.) iron.
- ⦿ Maximum solubility is 2.0 % C at 1130°C.
- ⦿ High formability, most of heat treatments begin with this single phase.
- ⦿ It is normally not stable at room temperature. But, under certain conditions it is possible to obtain austenite at room temperature.

Definition of structures

Austenite

⦿ Average properties are:

- Tensile strength = 150,000 psi;
- Elongation = 10 percent in 2 in.;
- Hardness = Rockwell C 40, approx; and
- toughness = high



Definition of structures

- ◎ **Cementite** or iron carbide, is **very hard, brittle** intermetallic compound of iron & carbon, as Fe_3C , contains 6.67 % C.
- ◎ It is the hardest structure that appears on the diagram, exact melting point unknown.
- ◎ Its crystal structure is orthorhombic.
- ◎ It has
 - low tensile strength (approx. 5,000 psi), but
 - high compressive strength.

Definition of structures

- ◎ **Ledeburite** is the eutectic mixture of austenite and cementite.
- ◎ It contains 4.3 percent C and is formed at 1130°C.

Definition of structures

- ⑩ **Martensite** - a super-saturated solid solution of carbon in ferrite.
- ⑩ It is formed when steel is cooled so rapidly that the change from austenite to pearlite is suppressed.
- ⑩ The interstitial carbon atoms distort the BCC ferrite into a BC-tetragonal structure (BCT).; responsible for the hardness of quenched steel

The Iron-Iron Carbide Diagram

- ◎ A map of the temperature at which different phase changes occur on very slow heating and cooling in relation to Carbon, is called Iron- Carbon Diagram.
- ◎ Iron- Carbon diagram shows
 - the type of alloys formed under very slow cooling,
 - proper heat-treatment temperature and
 - how the properties of steels and cast irons can be radically changed by heat-treatment.

Various Features of Fe-C diagram

Phases present

L

α ferrite
BCC structure
Ferromagnetic
Fairly ductile

δ

BCC structure
Paramagnetic

γ austenite

FCC structure
Non-magnetic
ductile

Fe_3C cementite

Orthorhombic
Hard
brittle

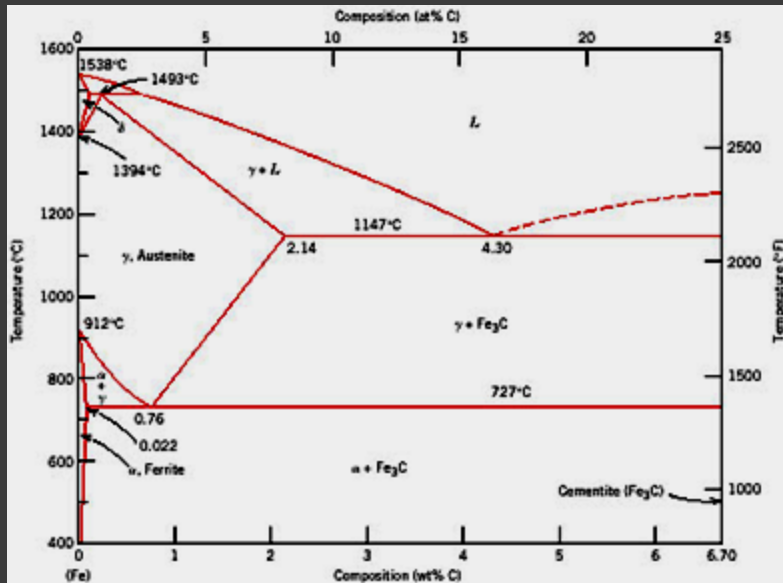


FIGURE 9.22 The iron-iron carbide phase diagram. (Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski, Editor-in-Chief, 1990. Reprinted by permission of ASM International, Materials Park, OH.)

Reactions

Peritectic $L + \delta = \gamma$

Eutectic $L = \gamma + \text{Fe}_3\text{C}$

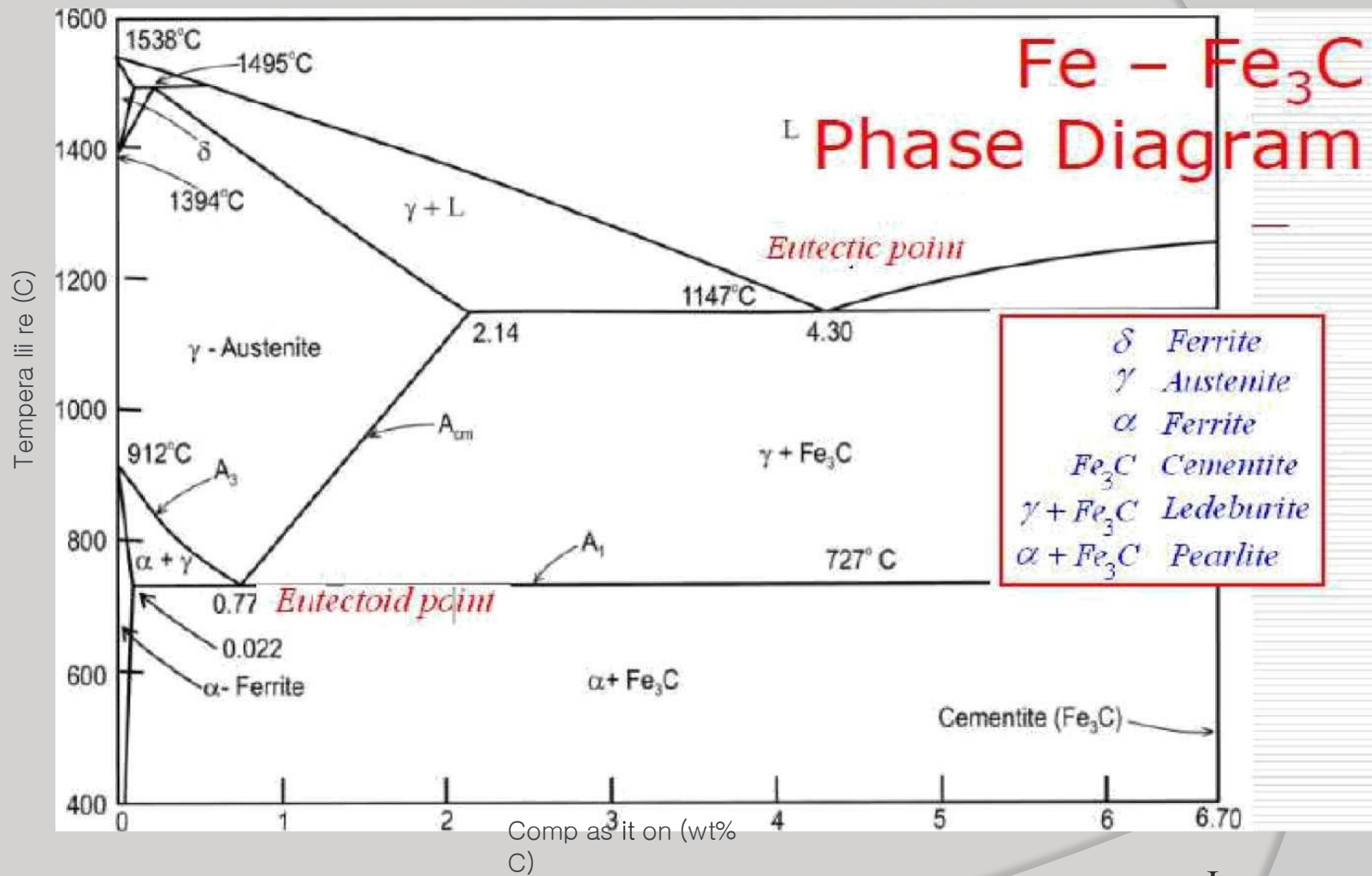
Eutectoid $\gamma = \alpha + \text{Fe}_3\text{C}$

Max. solubility of C in ferrite=0.022%

Max. solubility of C in austenite=2.11%

Three Phase Reactions

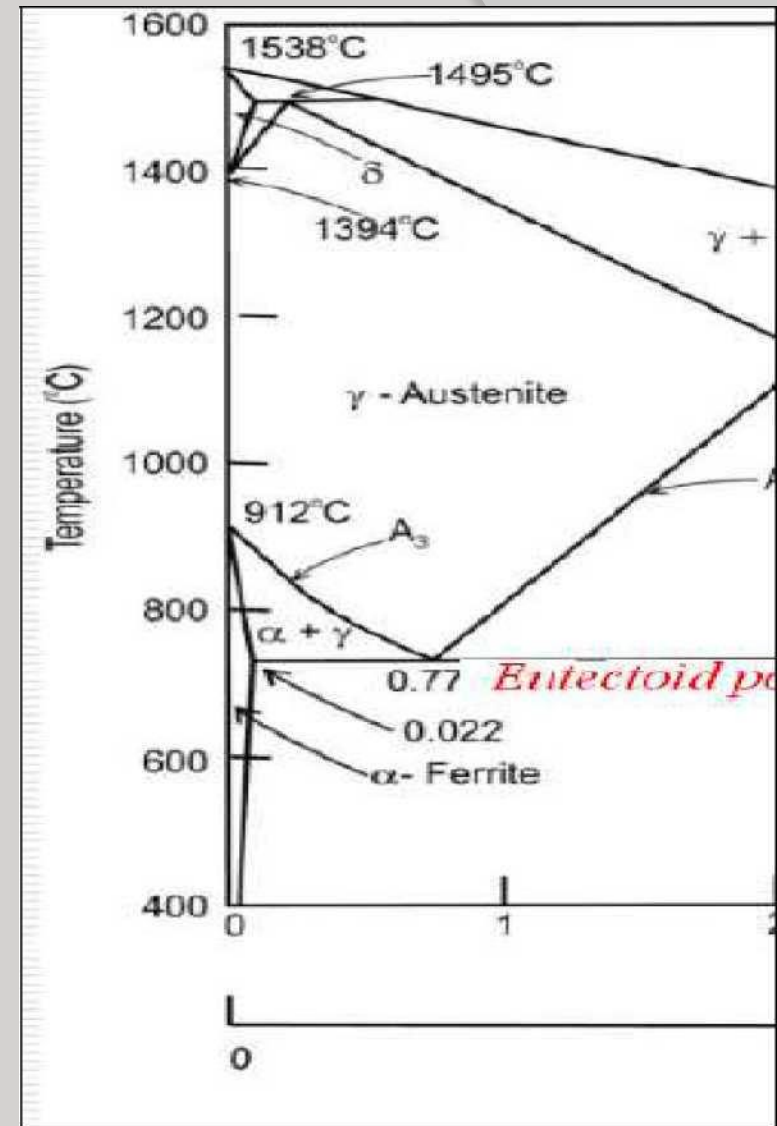
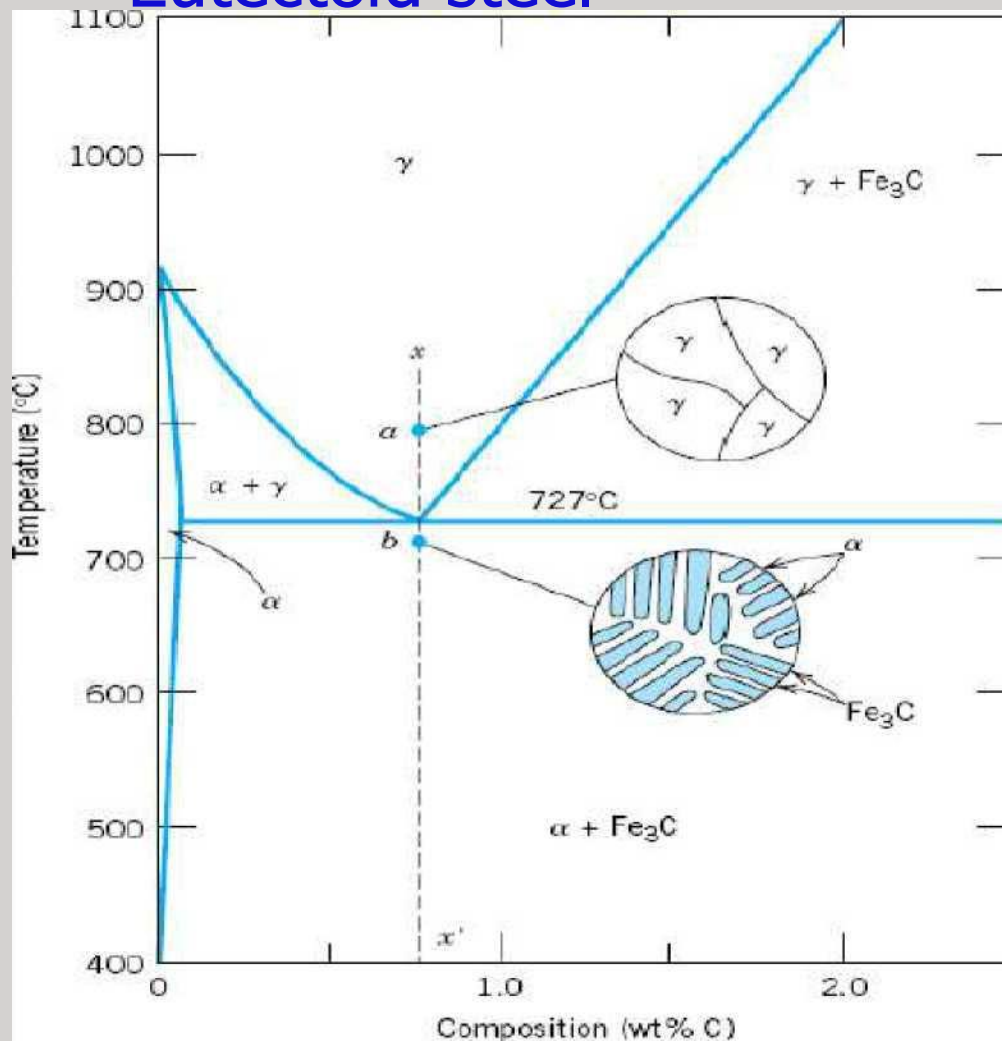
- ⦿ **Peritectic**, at 1490 deg.C, with low wt% C alloys (almost no engineering importance).
- ⦿ **Eutectic**, at 1130 deg.C, with 4.3wt% C, alloys called **cast irons**.
- ⦿ **Eutectoid**, at 723 deg.C with eutectoid composition of 0.8wt% C, two-phase mixture (ferrite & cementite). They are **steels**.



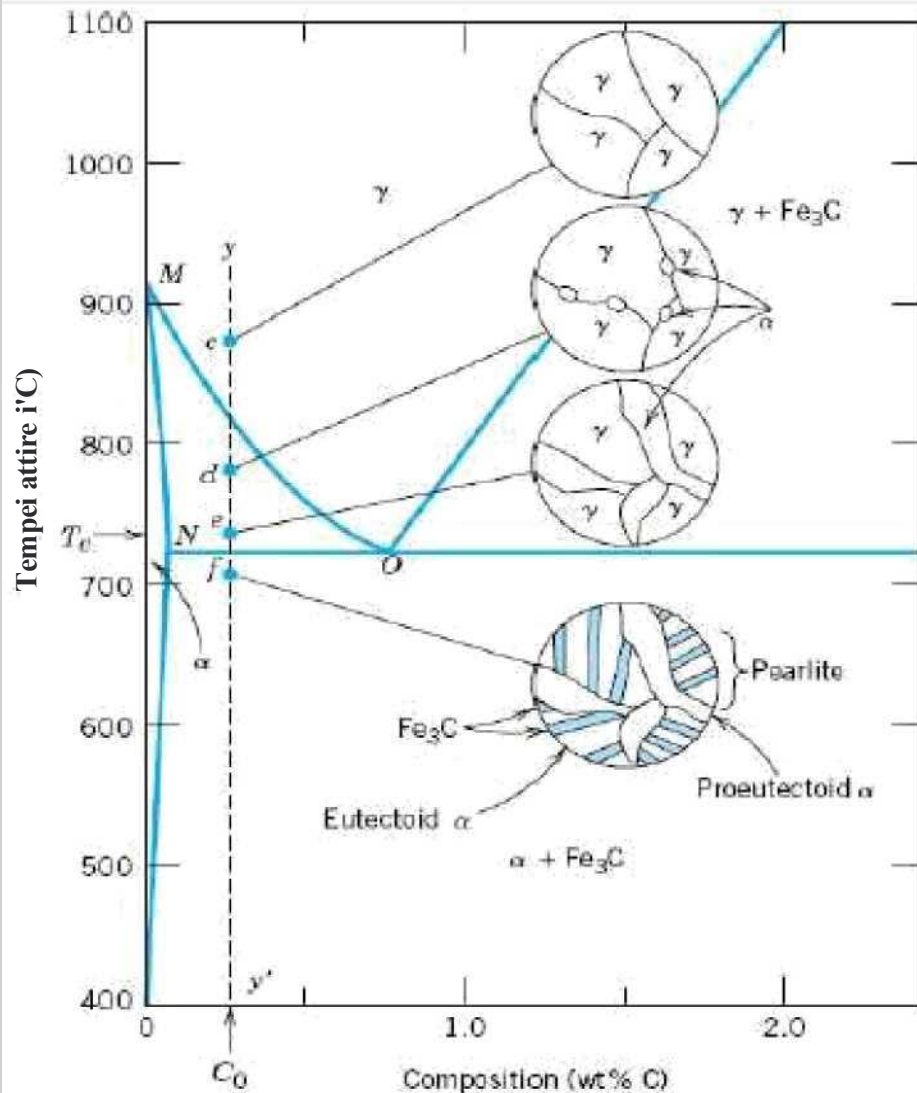
I 5a
Composition fw(V_c Fe C)

J -
too

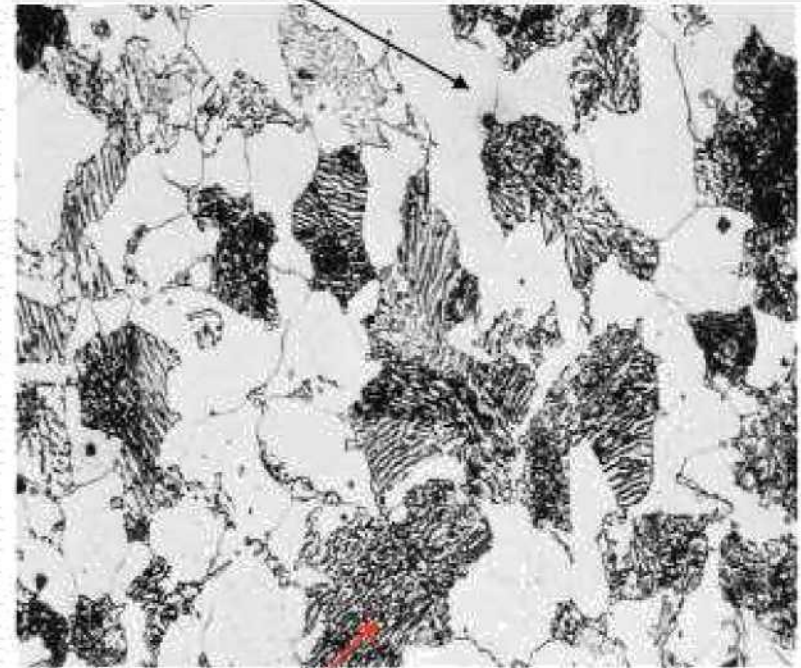
Eutectoid steel



From previous fig. of FE-C Diagram



Proeutectoid
Ferrite



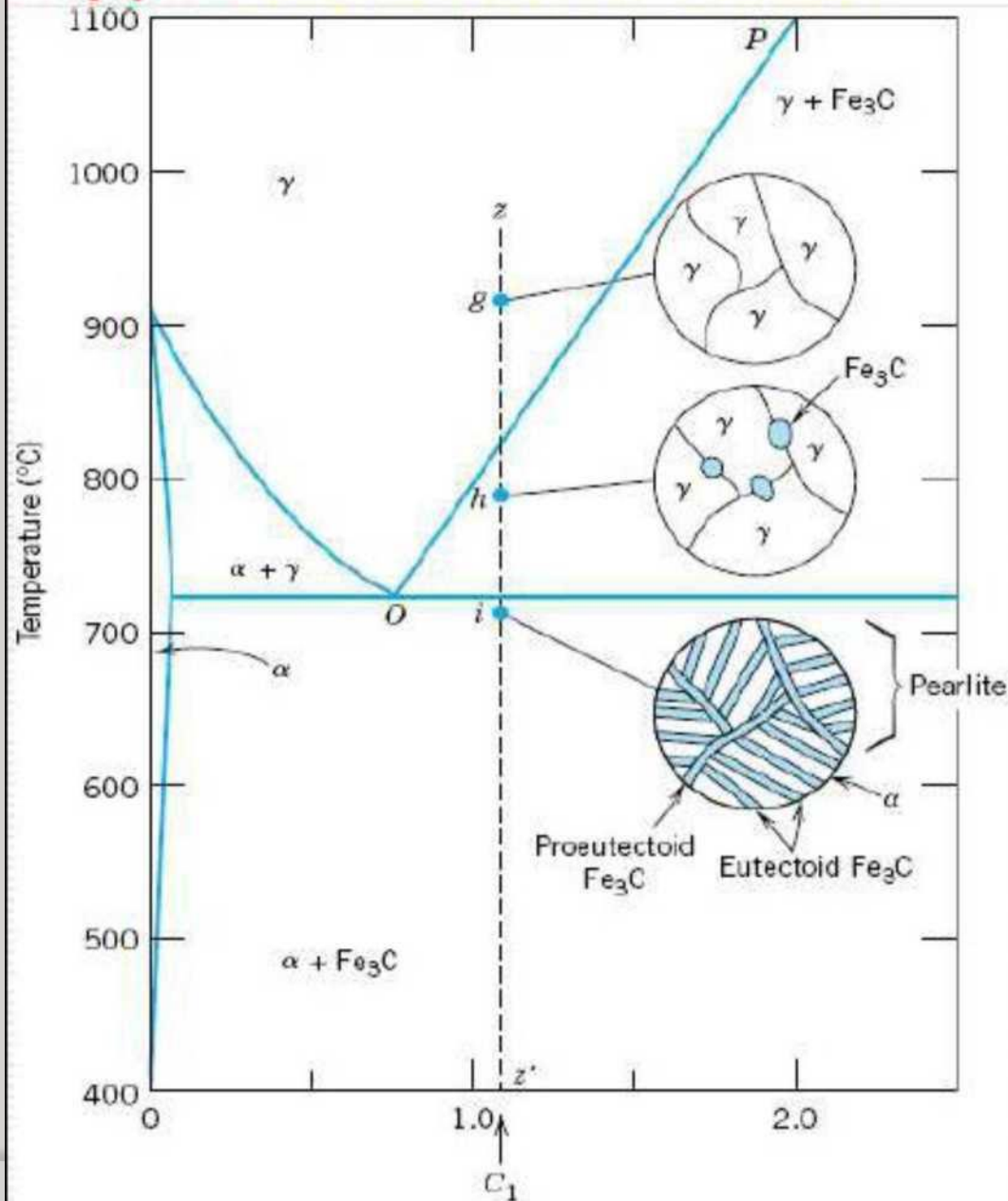
Pearlite

Microstructure of 0.38 wt% C

hypoeutectoid steel

- **Hypo-eutectoid steels:** Steels having less than 0.8% carbon are called *hypo-eutectoid steels* (*hypo* means "less than").
- Consider the cooling of a typical hypo-eutectoid alloy along line $y-y$.
- At high temperatures the material is entirely austenite.
- Upon cooling it enters a region where the stable phases are ferrite and austenite.
- The low-carbon ferrite nucleates and grows, leaving the remaining austenite richer in carbon.
- At 723°C , the remaining austenite will have assumed the eutectoid composition (0.8% carbon), and further cooling transforms it to pearlite.
- The resulting structure, is a mixture of *primary or proeutectoid ferrite* (ferrite that forms before the eutectoid reaction) and regions of pearlite.

Hypereutectoid steel



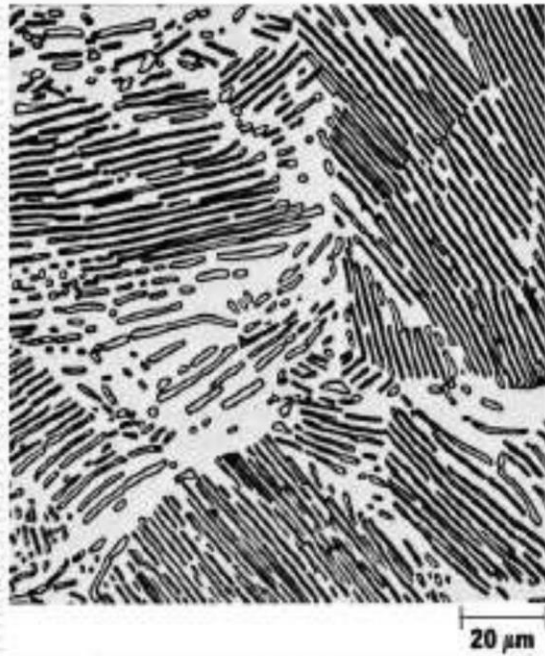
Pearlite



Proeutectoid
cementite

Microstructure of 1.4 wt% C
hypereutectoid steel

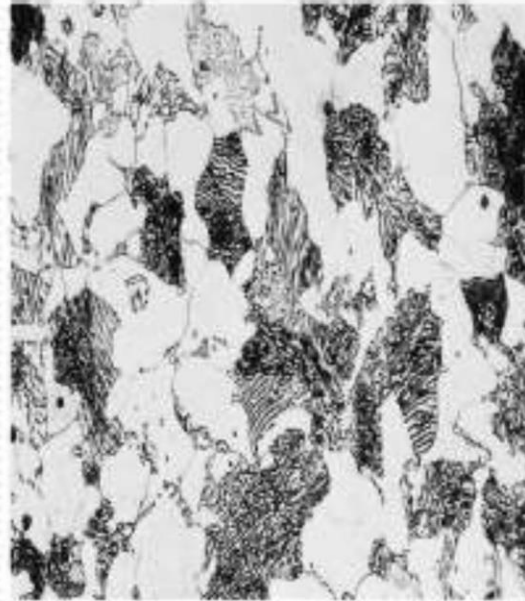
- **Hyper-eutectoid steels** (*hyper means* "greater than") are those that contain more than the eutectoid amount of Carbon.
- When such a steel cools, as along line $z-z'$, the process is similar to the hypo-eutectoid steel, except that the primary or pro-eutectoid phase is now cementite instead of ferrite.
- As the carbon-rich phase nucleates and grows, the remaining austenite decreases in carbon content, again reaching the eutectoid composition at 723°C.
- This austenite transforms to pearlite upon slow cooling through the eutectoid temperature.
- The resulting structure consists of primary cementite and pearlite.
- The continuous network of primary cementite will cause the material to be extremely brittle.



Eutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite



Hypoeutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite +
proeutectoid ferrite



Hypereutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite +
proeutectoid
cementite

Three invariant reactions

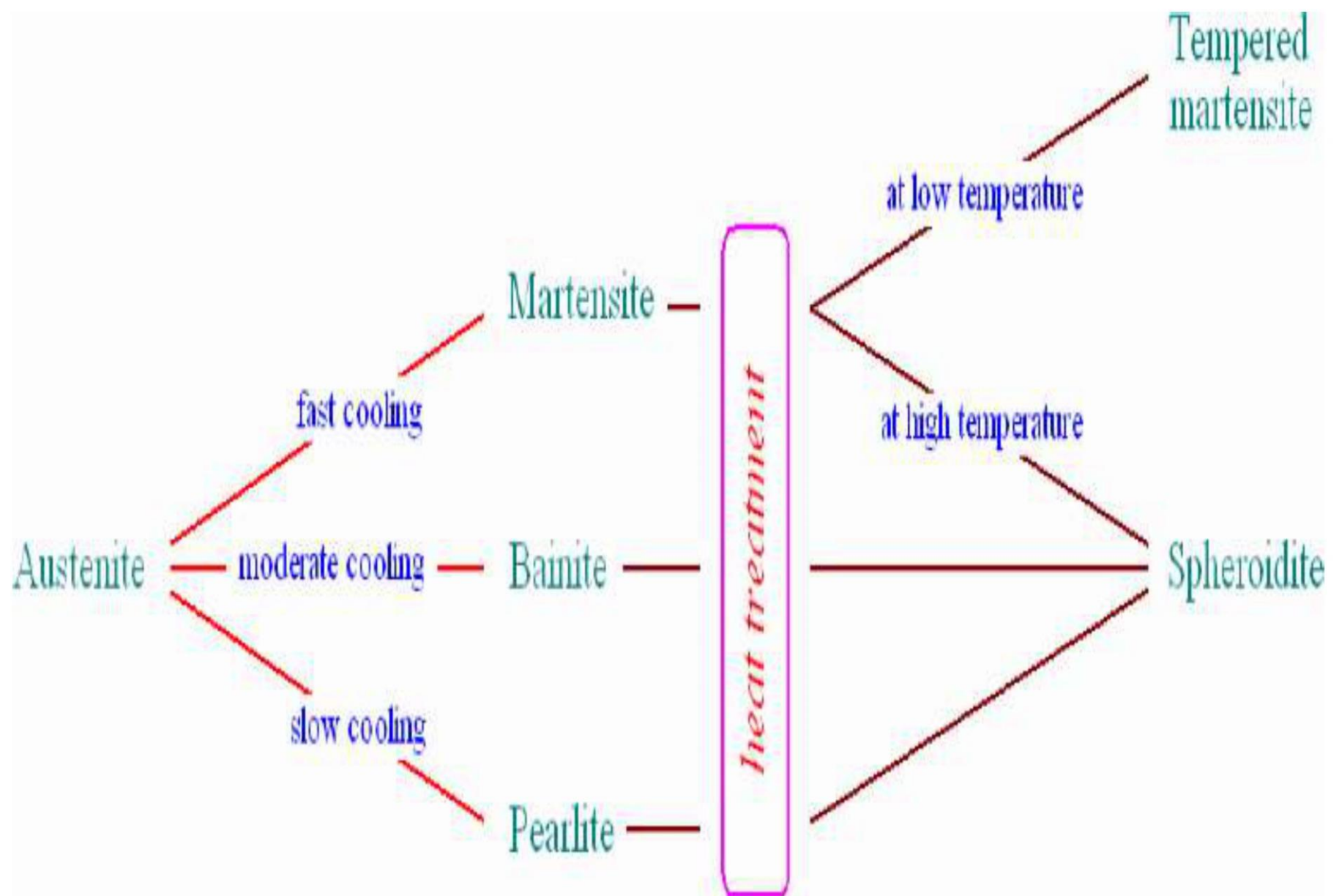
A horizontal line always indicates an invariant reaction in binary phase diagrams

- Peritectic reaction at 1495°C and $0.18\%\text{C}$,
 - $\delta\text{-ferrite} + L \leftrightarrow \gamma\text{-iron (austenite)}$
- Eutectic reaction at 1147°C and $4.3\%\text{C}$,
 - $L \leftrightarrow \gamma\text{-iron} + \text{Fe}_3\text{C (cementite) [ledeburite]}$
- Eutectoid reaction at 727°C and $0.77\%\text{C}$,
 - $\gamma\text{-iron} \leftrightarrow \alpha\text{-ferrite} + \text{Fe}_3\text{C (cementite) [pearlite]}$

DEFINITION OF STRUCTURES

Various phases that appear on the **Iron-Carbon equilibrium phase diagram** are as under:

- Austenite
- Ferrite
- Pearlite
- Cementite
- Martensite*
- Ledeburite



Austenite

Austenite is an interstitial solid solution of Carbon dissolved in F.C.C. iron.

Maximum solubility is 2.0 % C at 1130°C.

High formability, most of heat treatments begin with this single phase. It is normally not stable at room temperature. But, under certain conditions it is possible to obtain austenite at room temperature.

The Austenite to ferrite / cementite transformation in relation to Fe-C diagram

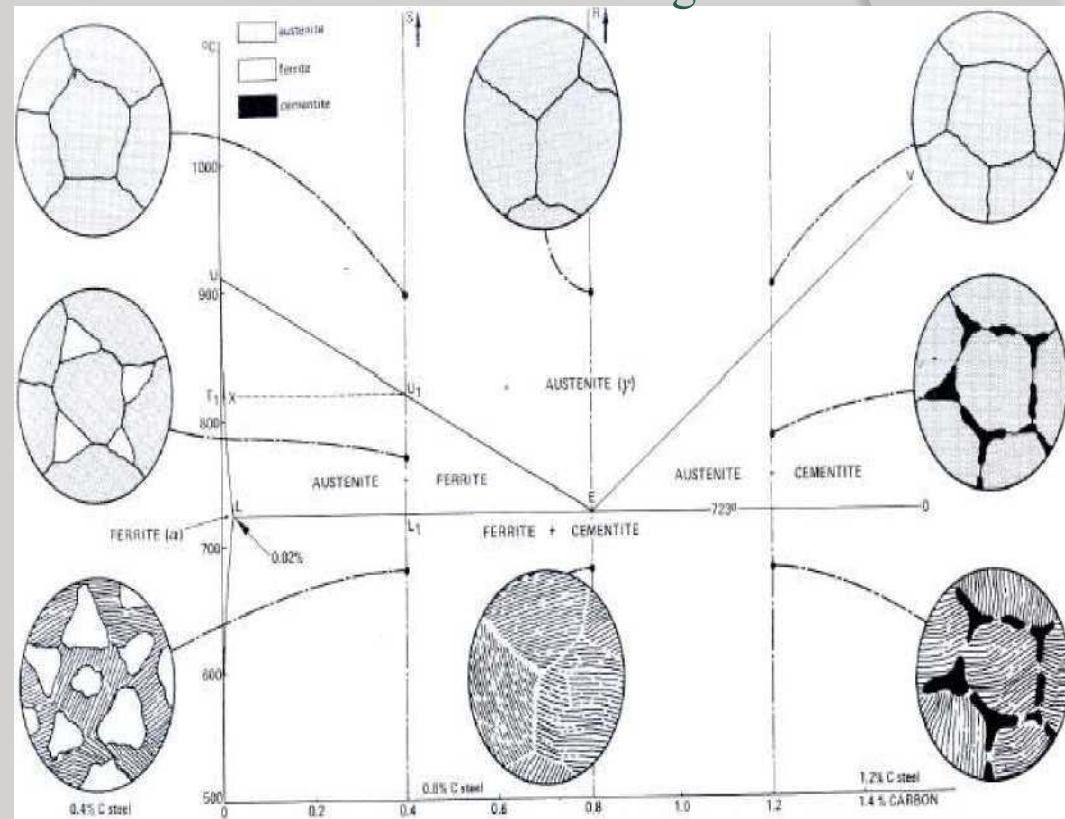


Fig 9.3 The austenite - ferrite / cementite transformation in relation to the iron carbon diagram.

Ferrite

- Ferrite is known as a solid solution.
- It is an interstitial solid solution of a small amount of carbon dissolved in a (BCC) iron. stable form of iron below 912 °C.
- The maximum solubility is 0.025 % C at 723°C and it dissolves only 0.008 % C at room temperature.
- It is the softest structure that appears on the diagram.

Pearlite

- Pearlite is the eutectoid mixture containing 0.80 % C and is formed at 723°C on very slow cooling.
- It is a very fine plate like or lamellar mixture of ferrite and cementite.
- The white ferrite background or matrix contains thin plates of cementite (dark).

Cementite or iron carbide

- Cementite or iron carbide, is very hard, brittle intermetallic compound of iron & carbon, as Fe_3C , contains 6.67 % C.
- It is the hardest structure that appears on the diagram, exact melting point unknown. Its crystal structure is orthorhombic. It has low tensile strength (approx. 5,000 psi), but high compressive strength.

Ledeburite

- Ledeburite is the eutectic mixture of austenite and cementite. It contains 4.3 percent C and is formed at 1130°C .

Martensite

- Martensite - a super-saturated solid solution of carbon in ferrite. It is formed when steel is cooled so rapidly that the change from austenite to pearlite is suppressed.
- The interstitial carbon atoms distort the BCC ferrite into a BC-tetragonal structure (BCT).; responsible for the hardness of quenched steel.

TTT DIAGRAM

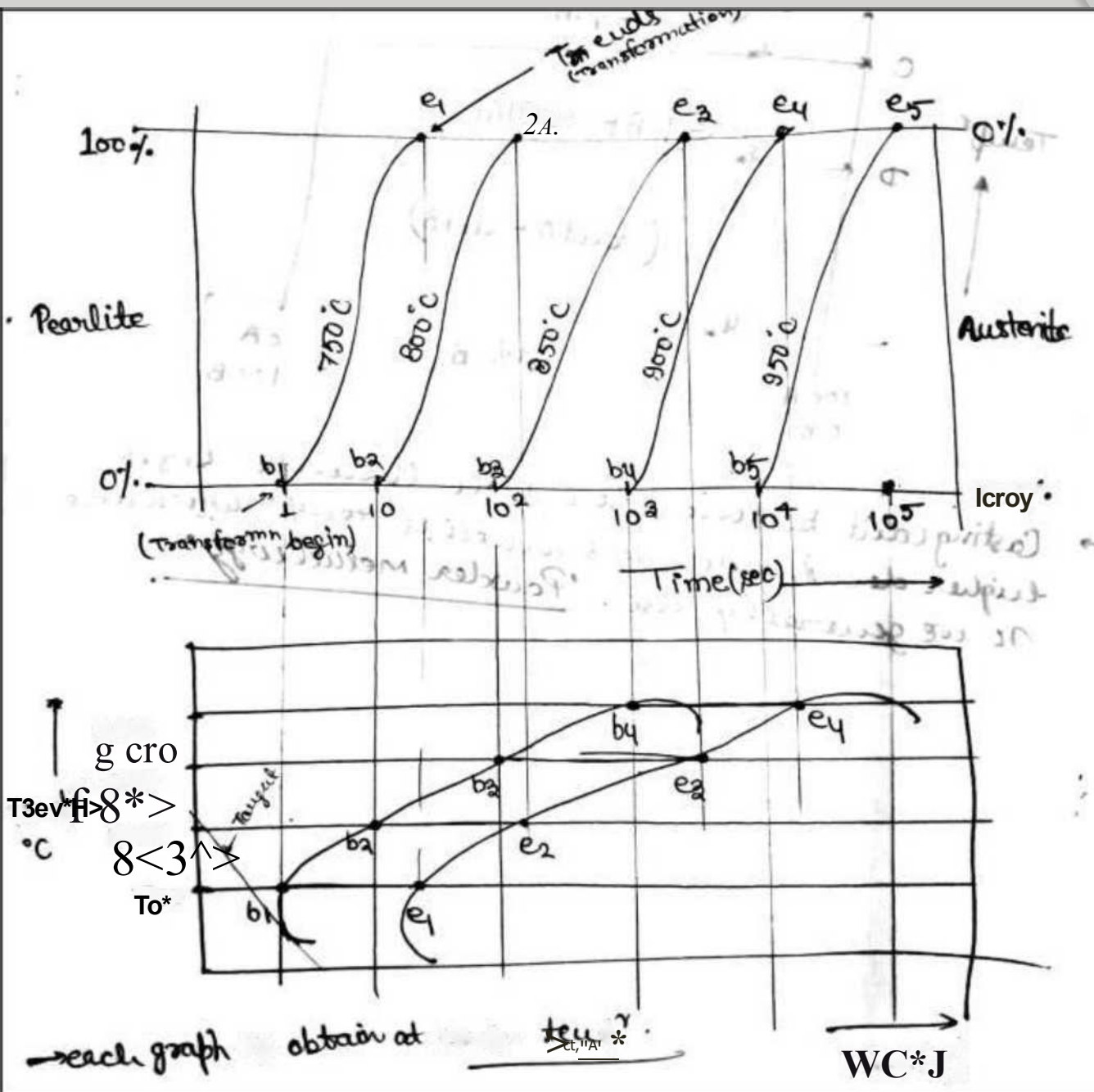
- T.T.T. shows relation between temperature & time taken for decomposition transformations to take place in a metal when the transformation is isothermal.
- Assess decomposition of austenite in a heat treatable steel.
- Provides information for the process of austenite decomposition under non-equilibrium conditions. (Transformation of austenite to the time & temperature conditions.)

DIFFERENCE BETWEEN IRON-CARBON & TTT

- Study of Fe-C diagram shows study of cooled steels under non equilibrium conditions.
- Doesn't involve **reaction condition** during heat treatment of steel.
- It only shows **phases & resulting microstructure** corresponding to equilibrium conditions.
- Fixing to **austenitizing temperature & predicting phases eventually obtained at given % of C & temperature.**
- Microstructure & properties of steel depends upon rate of cooling.
- As **cooling rate increases** transformation temperatures are **lowered & metastable (non-equilibrium) phases** are formed.
- At a very high rate of cooling of steel produces **Martensite (nonequilibrium phase)**

STEPS TO CONSTRUCT TTT DIAGRAM

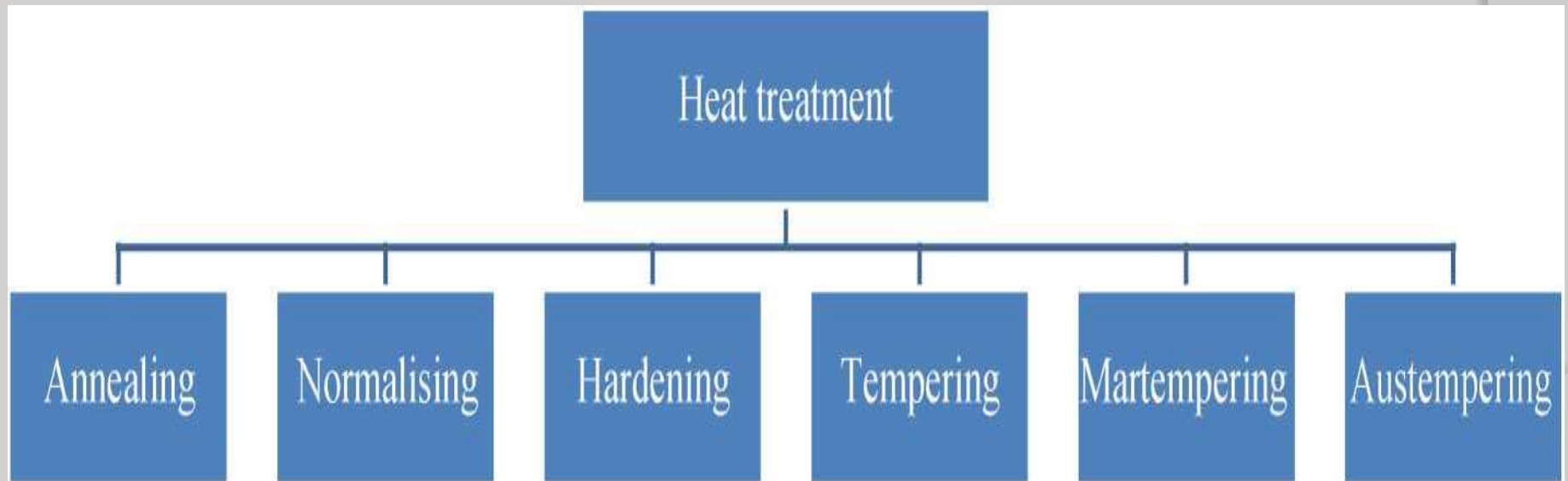
- Obtain large number of relatively small specimens. Place the sample in a molten salt bath held at the austenitizing temperature of 1080°C . Specimen are kept in a salt bath for a long period of time to form complete austenite.
- When austenitized, specimen is transferred to other salt bath at temperature of 810°C .
- After specimen react isothermally, quenched in cold water/ iced brine.
- As the specimen is quenched in cooled water, isothermal reaction stops & remaining austenite suddenly transforms into martensite.
- Reaction curve forms when large no. of specimen isothermally reacted for varying time periods.
- Finally **data obtained from a series of isothermal reaction curves (TTT)** for the whole temperature range of austenite instability for a given composition of steel.



- The transformation rate is inversely proportional to time at any temperature.
- Tangent to C-curves (TTT Curves) gives us the cooling rate.
- Based on the above curves the time required to transformation of Austenite-pearlite can be determined.
- TTT diagram denotes that it is drawn at different cooling rates, hence it is not recorded as an equilibrium diagram as Iron carbide diagram.

HEAT TREATMENT

- An operation or combination of operations which involves heating & cooling of a metal/alloy in solid state to obtain desirable conditions & properties.



PURPOSE OF HEAT TREATMENT

Heat treatment is carried out to

- (1) Cause relief of internal stresses developed during cold working, welding, casting, forging etc.
- (2) Harden & strengthen metals
- (3) Improve machinability
- (4) Change grain Size
- (5) Soften metals for further working as in wire drawing or cold rolling
- (6) Improve ductility & toughness
- (7) Increase heat, wear & corrosion resistance of materials
- (8) Improve electrical & magnetic properties
- (9) Homogenize the structure

Heat treatment techniques include

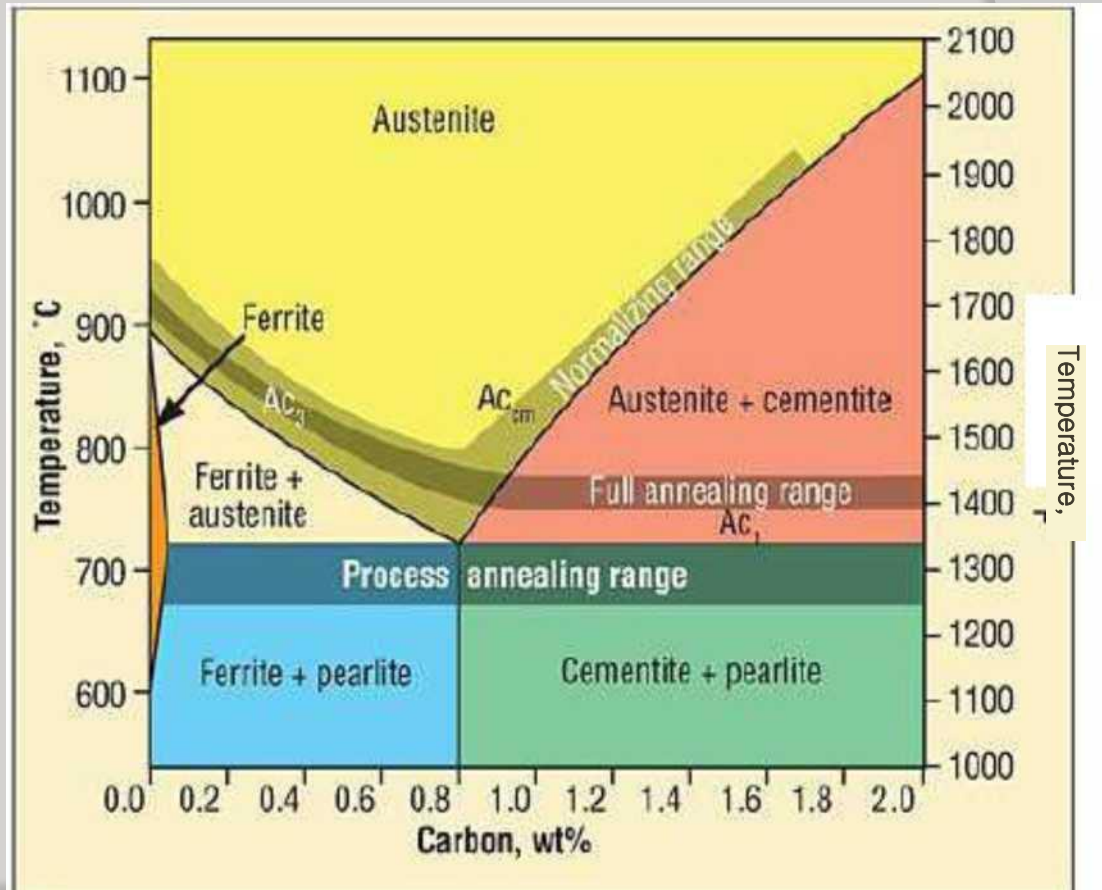
Annealing,
Hardening,
Tempering,
Quenching.

Annealing: Process Annealing

In this treatment, steel (or any material) is heated to a temperature below the lower critical temperature, and is held at this temperature for sufficient time and then cooled.

Cooling rate is of little importance as the process is being done at sub critical temperatures.

The purpose of this treatment is to reduce hardness and to increase ductility of cold-worked steel so that further working may be carried easily.



Annealing: Stress Relieving

As the name suggests, this process is employed to relieve internal stresses. No microstructural changes occur during the process.

Internal stresses are those stresses which can exist within a body in the absence of external forces. These are also known as residual stresses are locked-in stresses.

These stresses are developed in operations like:

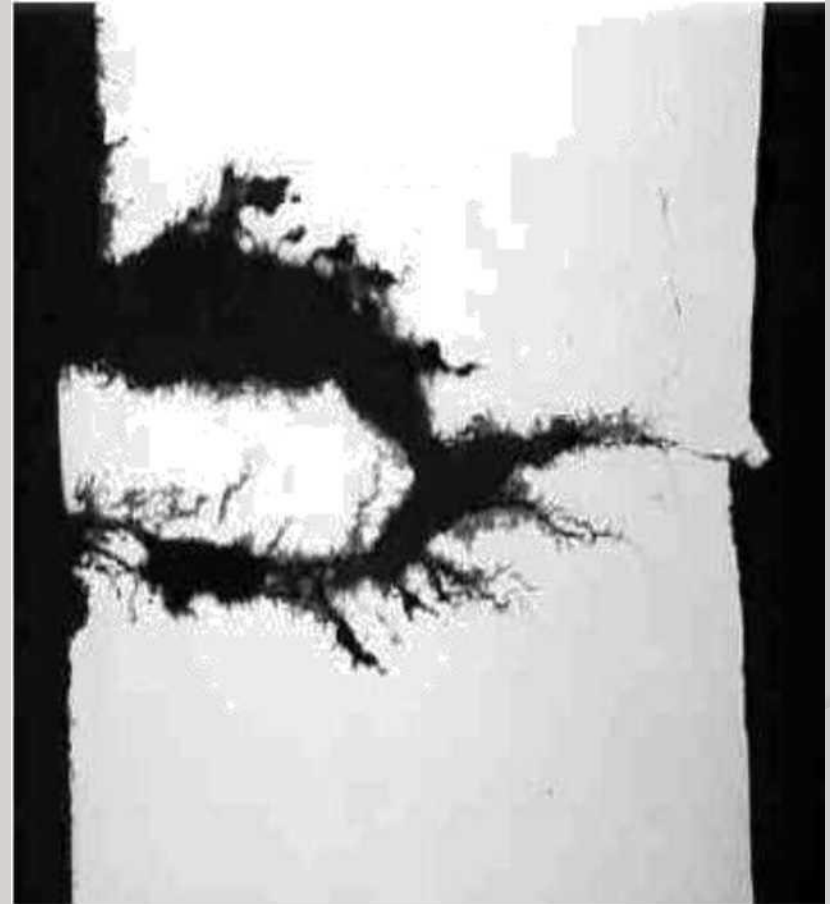
Solidification of castings, welding, machining, grinding, shot peening, surface hammering, cold working, case hardening, electroplated coatings, precipitation and phase transformation.

These internal stresses under certain conditions can have adverse effects: example: Steels with residual stresses under corrosive environment fail with stress corrosion cracking.

These stresses also enhance the tendency of steels towards warpage and dimensional instability.

Fatigue strength is reduced considerably when residual tensile stresses are present in steel.

The problems associated with internal stresses are more difficult in brittle materials than in ductile materials.



Stress - Corrosion Cracking

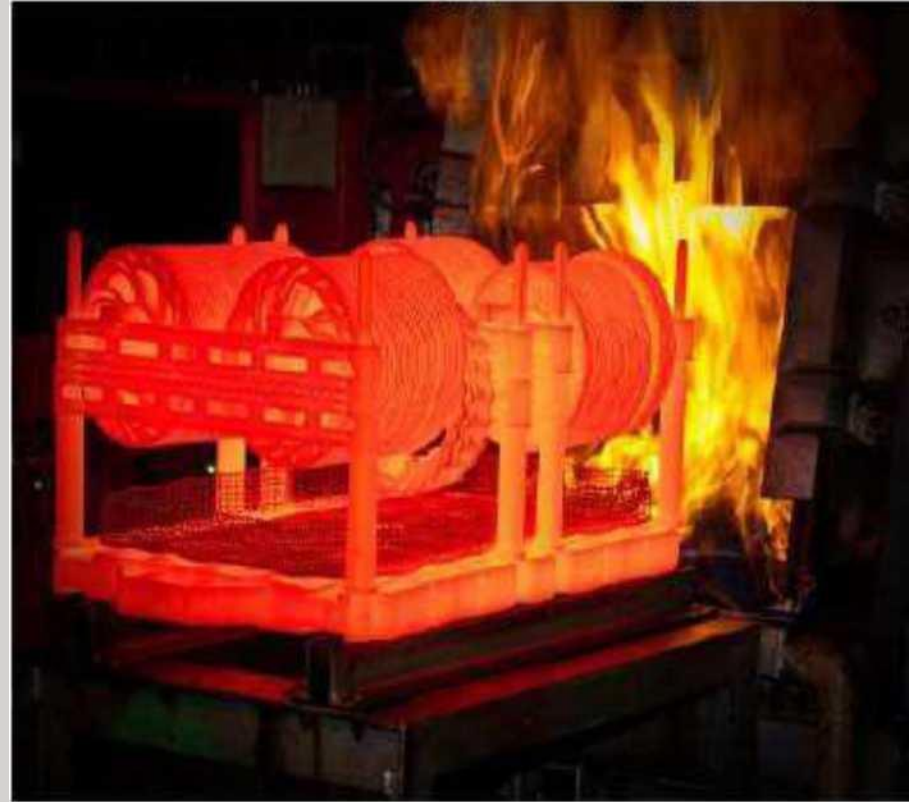
The process of stress relieving consists of heating materials uniformly to a temperature below the lower critical temperature, holding at this temperature for sufficient time, followed by uniform cooling

Uniform cooling is of utmost importance as non-uniform cooling will itself result in the development of internal stresses. Thus the main purpose of stress relieving will be lost.

Plain carbon steels and low alloy steels generally temperature is limited to 600 °C. Higher temperature is used for high alloy steels.

s J

The extent of the stresses relieved depend upon the temperature employed.

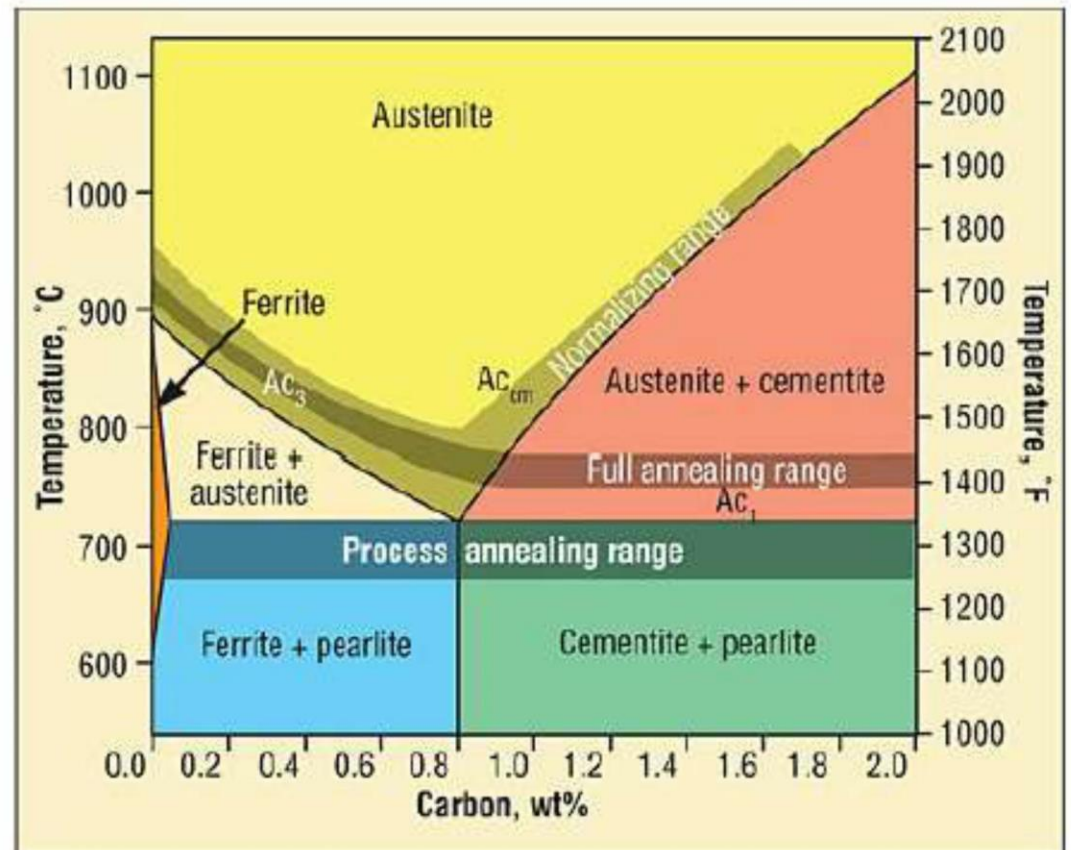


Annealing: Normalizing

➤ Normalizing is similar to full annealing, except steel is generally cooled in still air.

➤ The normalizing consists of heating steel to about 40-55 °C above critical temperature (Ac_3 or Ac_{cm}), and holding for proper item and then cooling in still air or slightly agitated air to room temperature.

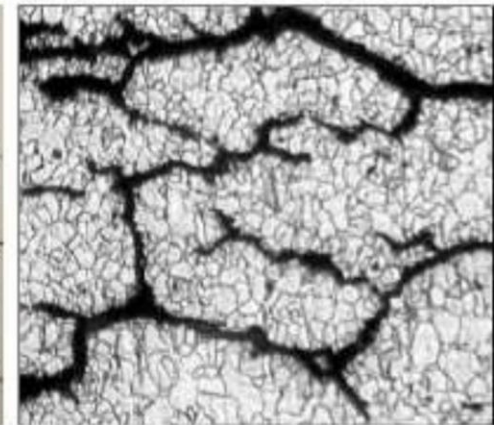
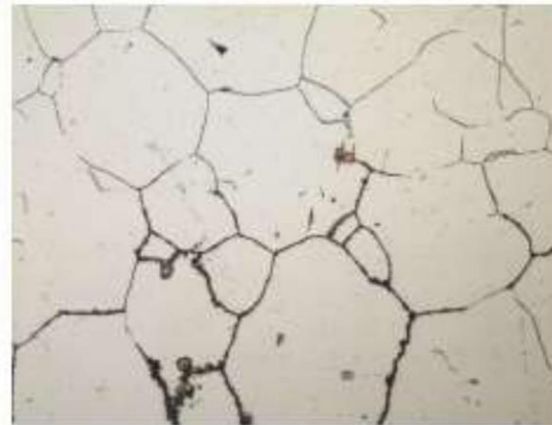
➤ In some special cases, cooling rates can be controlled by either changing air temperature or air volume.



➤ Normalized steels are generally stronger and harder than fully annealed steels.

➤ Steels are soft in annealed condition and tend to stick during machining. By normalizing, an optimum combination of strength and softness is achieved, which results in satisfactory level of Machinability in steels.

➤ Normalizing is the effective way to eliminate the carbide network.



➤ Normalized treatment is frequently applied to steel in order to achieve any one or more of the objectives, namely:

- To refine the grain structure,
- To obtain uniform structure,
- To decrease residual stresses,
- To improve Machinability.



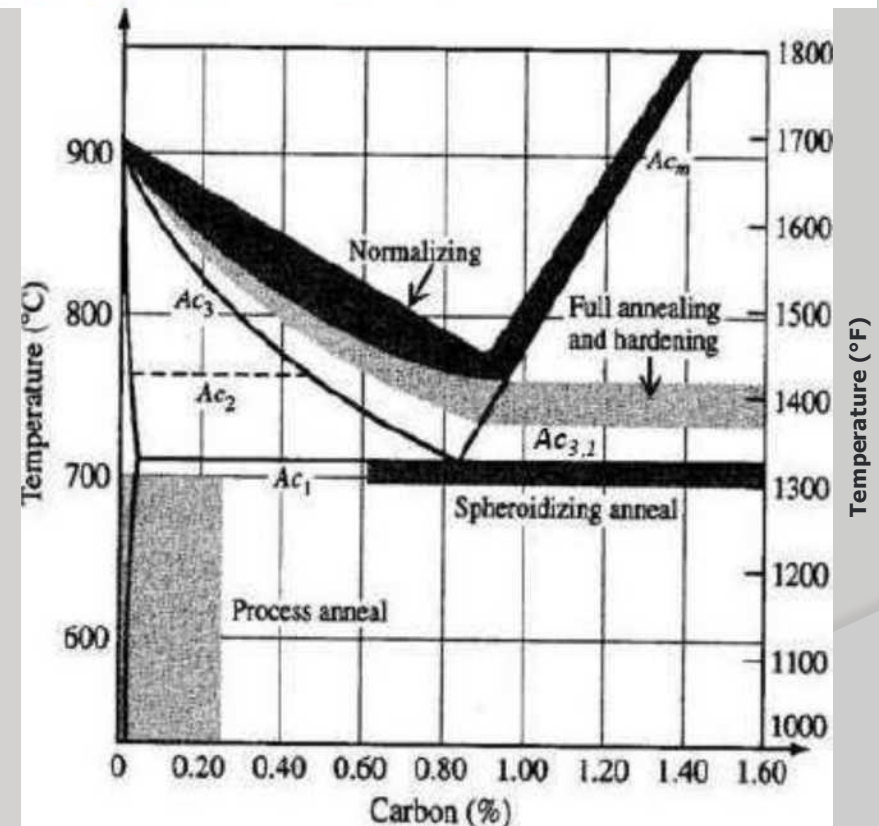
Hardening

Hardening and Hardness are two very different things. One is a process of heat treatment and other is an extrinsic property of a material.

Hardening is a heat treatment process in which steel is rapidly cooled from austenitising temperature. As a result to hardening, the hardness and wear resistance of steel are improved.

Hardening treatment generally consists of heating to hardening

temperature, holding at that temperature, followed by rapid cooling such as quenching in oil or water or salt baths.



Temperature Range for Heat Treatment of Carbon Steels

The high hardness developed by this process is due to the phase transformation accompanying rapid cooling. Rapid cooling results in the transformation of austenite at considerably low temperature into non equilibrium products.

The hardening temperature depends on chemical composition. For plain carbon steels, it depends on the carbon content alone. Hypoeutectoid steels are heated to about 30 - 50 °C above the upper critical temperature, whereas eutectoid and hyper eutectoid steels are heated to about 30 - 50 °C above lower critical temperature.

Ferrite and pearlite transform to austenite at hardening temperature for hypoeutectoid steel. This austenite transforms to martensite on rapid quenching from hardening temperature. The presence of martensite accounts for high hardness of quenched steel.

Hardening is applied To cutting tools and machine pans where high hardness and wear resistance are important.

The Process Variables:

Hardening Temperature: The steel should be heat treated to optimum austenitising temperature. A lower temperature results lower hardness due to incomplete transformation austenite. If this temperature is too high will also results lower hardness due to a coarse grained structure.

Soaking Time: Soaking time at hardening temperature should be long enough to transform homogenous austenite structure. Soaking time increases with increase in section thickness and the amount of alloying element.

Delay in quenching: After soaking, the steel is immediately quenched. Delay in quenching may reduce hardness due to partial transformation of austenite.

Type of quenching medium also has a profound effect, which will be discussed briefly

The main purpose of hardening tool steel is to develop high hardness. This enables tool steel to cut other metals. High hardness developed by this process also improves wear resistance. Gears, shafts and bearings. Tensile strength and yield strength are improved considerably by hardening structural steels.

Because of rapid cooling, high internal stresses are developed in the hardened steel. Hence these steels are generally brittle.

Hardening in general is followed by another treatment known as tempering which reduces internal stresses and makes the hardened steel relatively stable.

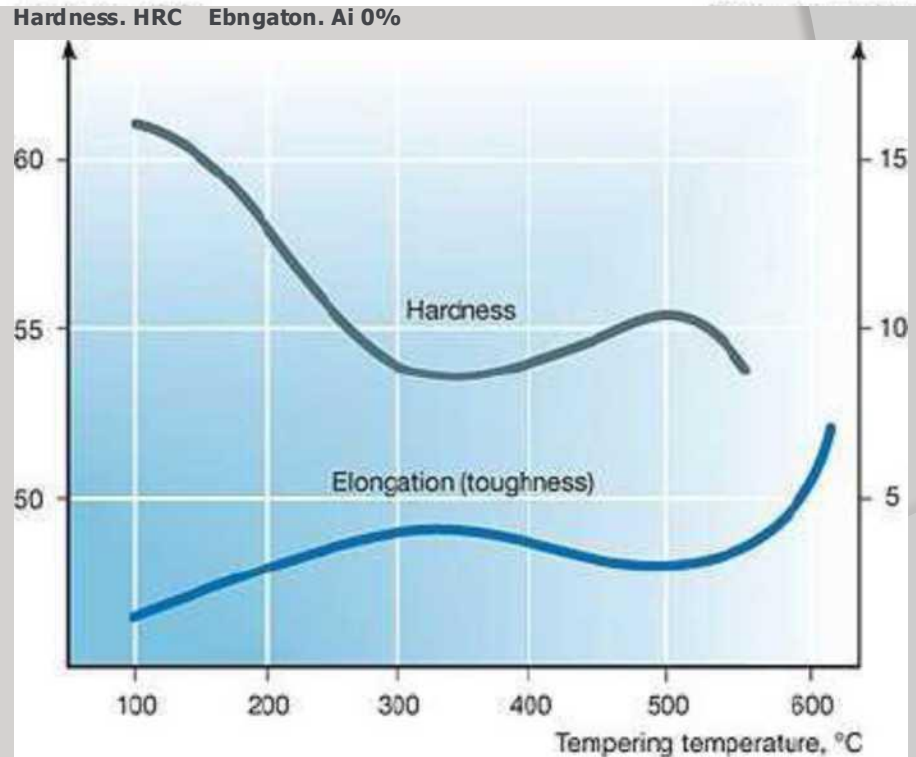


Tempering

Hardened steels are so brittle that even a small impact will cause fracture. Toughness of such a steel can be improved by tempering. However there is small reduction in strength and hardness.

Tempering is a sub-critical heat treatment process used to improve the toughness of hardened steel.

Tempering consists of reheating of hardened steel to a temperature below Lower critical temperature and is held for a period of time, and then slowly cooled in air to room temperature.

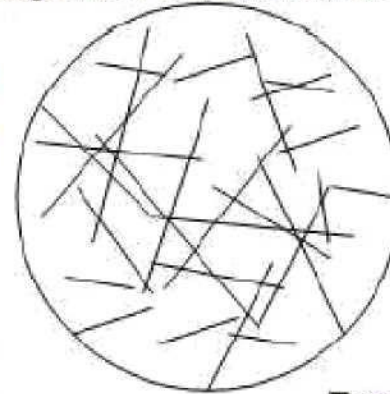


At tempering temperature, carbon atoms diffuse out and form fine cementite and softer ferrite structure left behind. Thus the structure of tempered steel consists of ferrite and fine cementite.

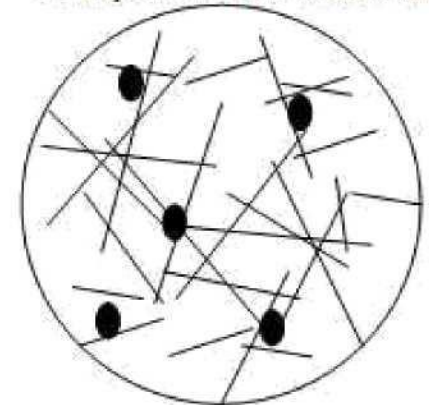
➤ Thus tempering allows to precipitate carbon as very fine carbide and allow the microstructure to return to BCC

➤ The temperatures are related to the function of the parts. Cutting tools are tempered between 230 – 300 °C. If greater ductility and toughness are desired as in case of shafts and high strength bolts, the steel is tempered in the range of 300 – 600 °C.

Toughened Microstructure



Tempered Microstructure



THANK YOU